ASML: Self-Adjusting Photolithography Lens

By: Michael Horowitz, Michael Moskowitz, Jacob Goldsmith, Joseph Picard, Jacob Rabinowitz

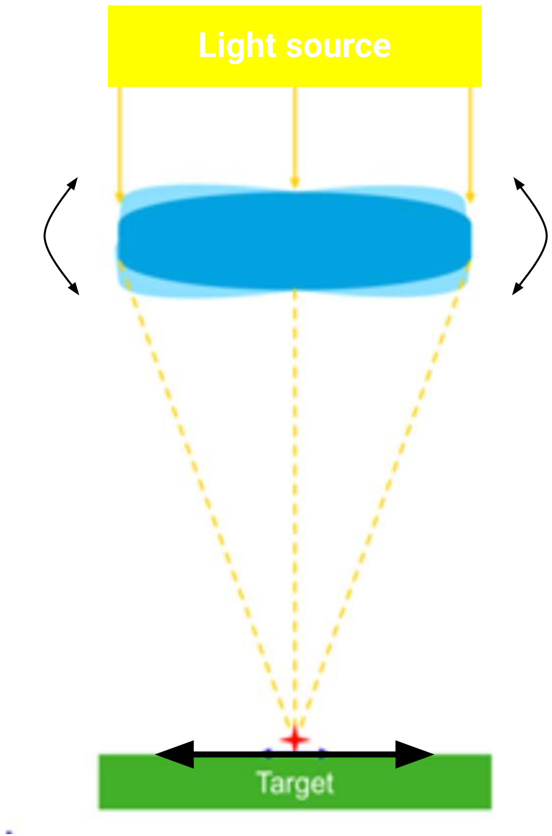
Contents

- Project Overview
- Key Design Requirements and Parameters
- Radially Compliant Mechanism and Approach
- Actuation Design
- Engineering Solutions
- Project Status and Next Steps

Problem statement

- A mechanical system to tilt an optic lens along two axis to:
  - shift a focal point

- Self-adjusting to account for:
  - vibration
  - thermal disturbances



Requirements and Parameters

Requirements

- Lens-tilting capacity: in the range of +/- 10 degrees in the Rx and Ry directions
- Movements in-plane: should have a 1000:1 ratio to lateral movements between planes

Design Parameters

- Lens optic: is specified between 4-8” Outer Diameter
- Mounting ring: between a 9” Inner diameter and a 12” OD
- Lens positioning: ideally occur in under 3 seconds

Radially Compliant Mechanism

Problem: Stress in the Optic Lens can cause Distortion, Focus Shift, or Fracture

The optic and the optic mount are made of materials which expand differently.

- Aluminum has a coefficient of thermal expansion of 23×10^{-6} /^{o}C
- Glass has a coefficient of thermal expansion of 9×10^{-6} /^{o}C

Solution: Radial Compliance

The optic is fixed in a non-rigid fixture that will reduce distortion.

- Allows motion in the radial direction
- Absorbs thermal expansion mismatch
- Maintains uniform contact pressure

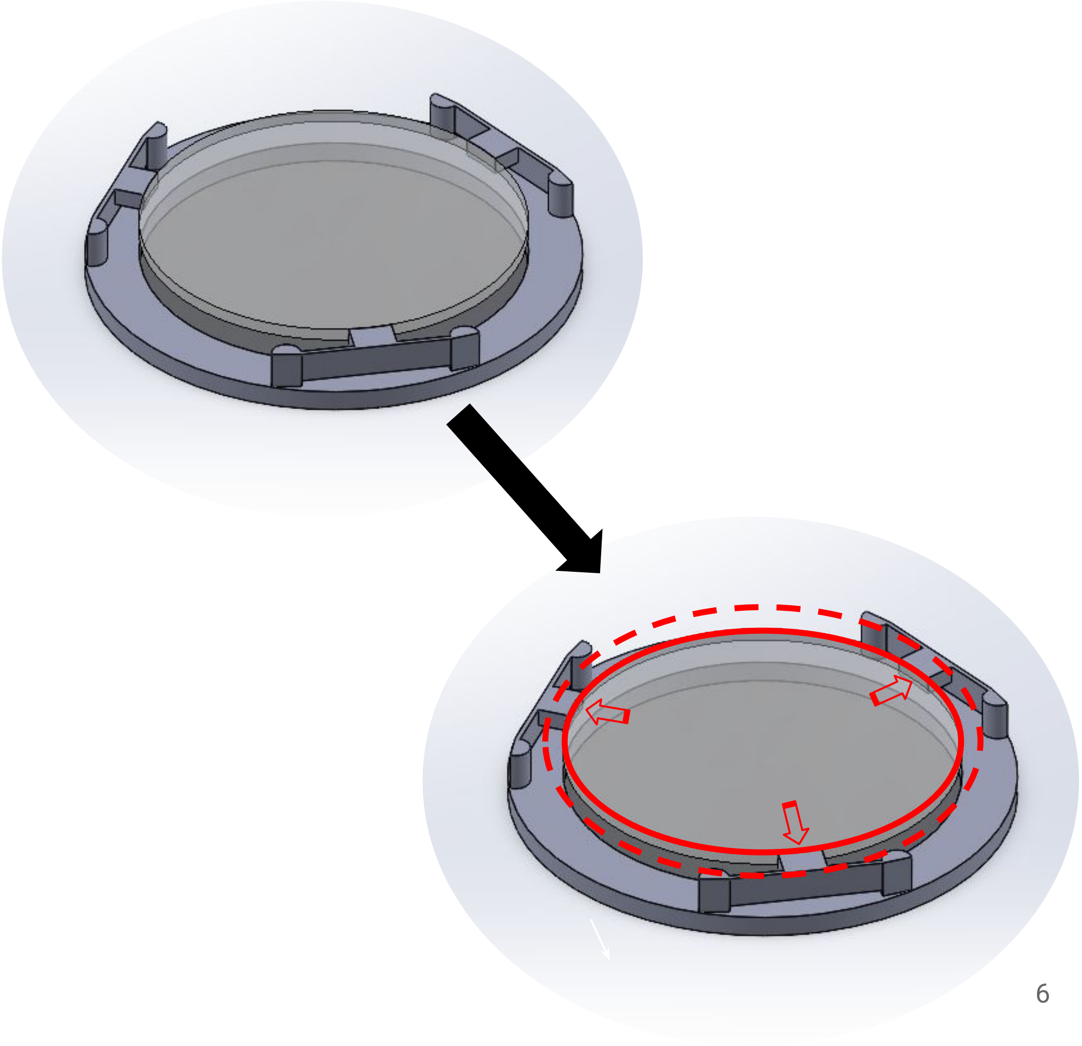
Approach:
The Edge-Bonded Flexure

Constraint: (To solve the stress effect problem in the lens)

- Lens mount has to be 1000x stiffer in the Z direction than in X and Y directions

How this model complies:

- Very stiff in the Z direction due to the relatively high moment of inertia
- Will flex on the X-Y plane due to to the thin members
- Due to the symmetry of the model, there will be invariant thermal Expansion.



Actuation and Control

Our Goal with Actuation:

- To allow for precise control of the lens orientation so that the light is accurately aimed at a fixed target.

Actuation: Converts an electrical signal into mechanical motion (force/displacement) to control the system.

Actuators: Devices that create motion, categorized as:

1) **Linear:** Straight-line (push/pull)

2) **Rotary:** Turning/spinning (tilt)

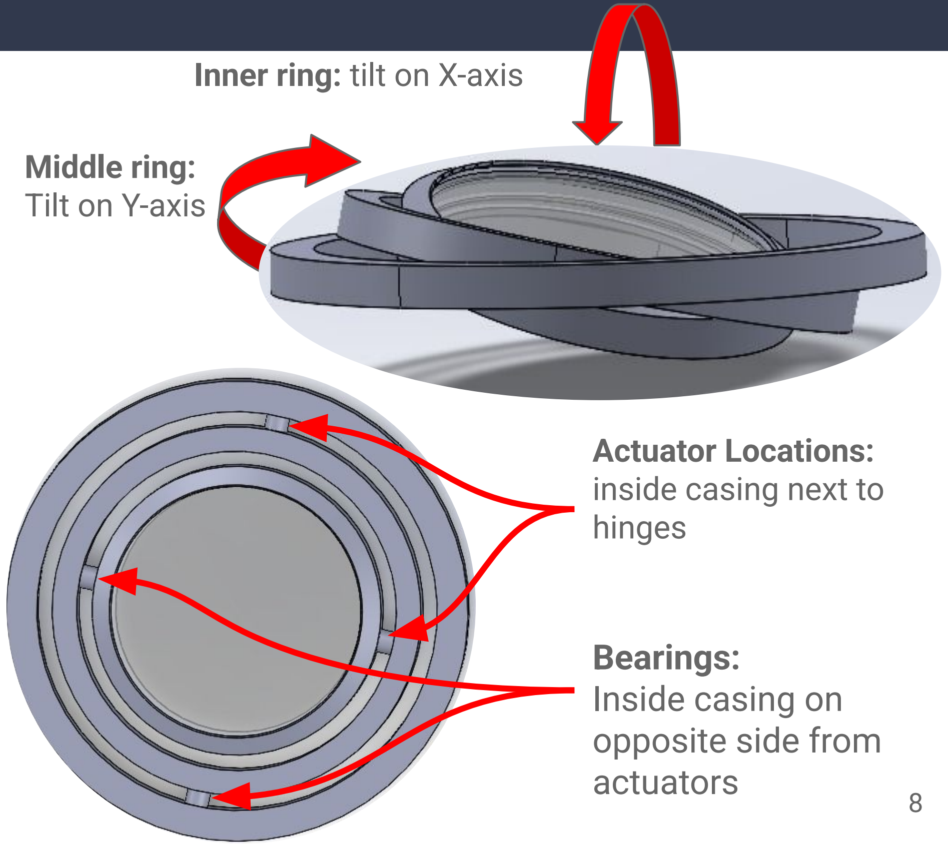
Approach A: Gyremy

How it works

- Uses two actuators to rotate lens about the X and Y Axis.
- Lens can focus onto a target on the entire plane below

Advantages

- Coordination between the actuators is very straight forward:
  - one is responsible for the X-angle adjustment, while the other manages the Y-angle.



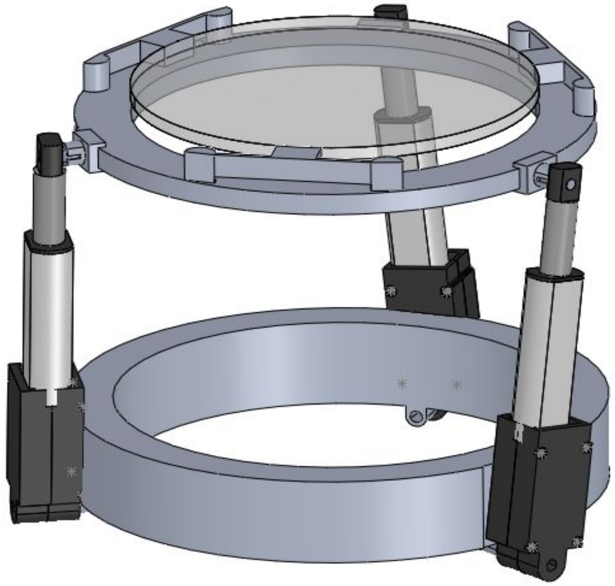
Approach B: Lynn

How It Works

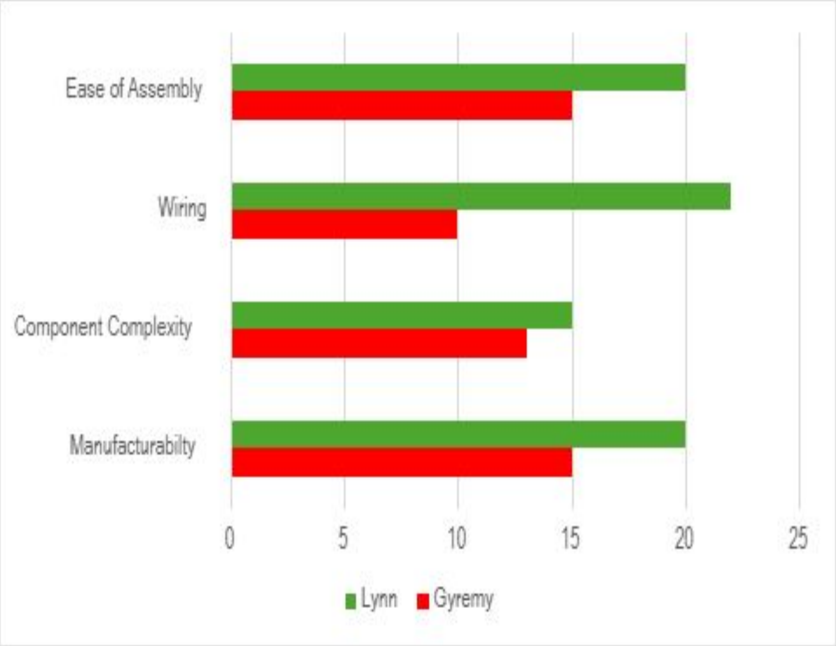
- Three linear actuators placed evenly around the lens extend or retract to control tilt and height.
- Tilting occurs by varying actuator lengths.

Advantages

- Precise and repeatable control of tilt and height
- Compact, simple, and low-maintenance design
- Easy to model and assemble



Ultimate Decision: Lynn

Functional requirement	**Approach 1: Gyremmy**	**Approach 2: Lynn**	**Reason why Linear system is better:**
Easy of Assembly			Lens casing is easier to access for Lynn
Wiring			For Gyremy, the inner actuators tilt; wiring would need to move with them. For Lynn, Wiring is independent from movement
Component complexity			Gyremy requires bearings and rotating components, whereas Lynn only needs standard ball and socket joints for the legs
Manufacturability			Manufacturing is for more complex for Gyremy and may require custom parts

Goals and Objectives

What's been done:

- Brainstorming and concept development
- Concept iteration and evaluation
- Preliminary concept modeling
- Final Concept Selection

What's left:

- Develop detailed Solid models
- Conduct Finite Element analysis
- Component selection
- Build a prototype

End of Semester goal:

- Have a completed prototype
- Full project timeline

The floor is open to questions and suggestions!

Thank you!